

NYC Noise Complaints: Response-Time Analysis

Midterm project

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1 Research question

How do complaint type, borough, timing, and location relate to the time the New York Police Department (NYPD) takes to close NYC 311 noise complaints? The analysis focuses on complaints created from June 30 through July 6, 2024 and uses a two-hour closure threshold as the binary modeling outcome.

The source is NYC Open Data's [311 Service Requests from 2020 to Present](#). The repository includes a fixed CSV snapshot so the results below do not change when the live dataset is revised.

2 Data cleaning

```
raw = load_raw_data(DATA_PATH)
complaints, audit = clean_complaints(raw)

pd.DataFrame(
    {
        "Measure": [
            "Raw rows",
            "Duplicate unique keys removed",
            "Rows with invalid/missing dates removed",
            "Rows with negative response times removed",
            "Analysis rows",
        ],
        "Count": [
            audit.raw_rows,
            audit.duplicate_rows,
```

```

        audit.invalid_date_rows,
        audit.negative_response_rows,
        audit.analysis_rows,
    ],
}
)

```

	Measure	Count
0	Raw rows	15398
1	Duplicate unique keys removed	0
2	Rows with invalid/missing dates removed	0
3	Rows with negative response times removed	0
4	Analysis rows	15398

Column names are normalized to snake case, dates are parsed explicitly, and records without a valid creation/closure pair are excluded. `unique_key` is used for de-duplication. ZIP codes remain strings so leading zeroes cannot be lost. Unlike the earlier draft, this workflow makes no live geocoding calls and does not rely on the obsolete `uszipcode` database.

```

pd.DataFrame(
    {
        "Field": ["borough", "complaint_type", "incident_zip", "latitude", "longitude"],
        "Missing": [complaints[c].isna().sum() for c in [
            "borough", "complaint_type", "incident_zip", "latitude", "longitude"
        ]],
    }
)

```

	Field	Missing
0	borough	1
1	complaint_type	0
2	incident_zip	3
3	latitude	94
4	longitude	94

3 Descriptive analysis

```

overall = complaints["response_time_hours"].describe(
    percentiles=[0.25, 0.5, 0.75, 0.9, 0.95]
)
overall.to_frame("Response time (hours)").round(2)

```

	Response time (hours)
count	15398.00
mean	1.45
std	3.13
min	0.00
25%	0.35
50%	0.66
75%	1.37
90%	2.94

	Response time (hours)
95%	4.99
max	50.90

Response times are right-skewed, so medians and interquartile ranges are more representative than means. The figure caps the display at the 99th percentile; all valid rows remain in the statistics and model.

The median closure time is 0.66 hours, and 16.1% of complaints take at least two hours to close.

```
cap = complaints["response_time_hours"].quantile(0.99)
fig, ax = plt.subplots()
ax.hist(
    complaints.loc[complaints["response_time_hours"] <= cap, "response_time_hours"],
    bins=35,
    color="#28666e",
    edgecolor="white",
)
ax.axvline(2, color="#b23a48", linestyle="--", linewidth=2, label="Two-hour threshold")
ax.set(xlabel="Closure time (hours)", ylabel="Complaints")
ax.legend(frameon=False)
plt.show()
```

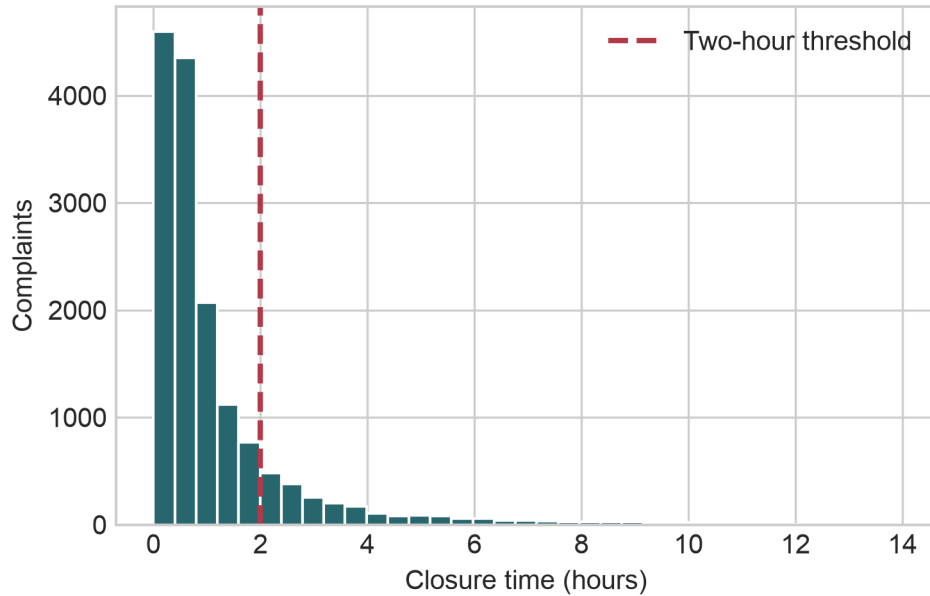


Figure 1: Distribution of NYPD noise-complaint closure times. The x-axis is limited to the 99th percentile for readability.

```
borough_summary = response_summary(complaints, "borough")
borough_summary.round({"median_hours": 2, "mean_hours": 2, "over_two_hour_rate": 3})
```

	borough	n	median_hours	mean_hours	over_two_hour_rate
0	BRONX	4406	0.88	1.69	0.235
1	QUEENS	3251	0.76	2.40	0.209
2	BROOKLYN	4093	0.55	0.96	0.103
3	MANHATTAN	3222	0.55	0.89	0.097

	borough	n	median_hours	mean_hours	over_two_hour_rate
4	STATEN ISLAND	425	0.50	0.77	0.075

```
complaint_summary = response_summary(complaints, "complaint_type")
complaint_summary.round({"median_hours": 2, "mean_hours": 2, "over_two_hour_rate": 3})
```

	complaint_type	n	median_hours	mean_hours	over_two_hour_rate
0	Noise - Residential	6895	0.70	1.77	0.188
1	Noise - Vehicle	1013	0.72	1.42	0.181
2	Noise - Commercial	1097	0.61	1.34	0.180
3	Noise - House of Worship	23	0.26	0.98	0.130
4	Noise - Street/Sidewalk	6147	0.62	1.14	0.126
5	Noise - Park	223	0.53	1.24	0.126

```
plot_data = borough_summary.sort_values("over_two_hour_rate")
rate = plot_data["over_two_hour_rate"]
se = (rate * (1 - rate) / plot_data["n"]).pow(0.5)

fig, ax = plt.subplots()
ax.barh(plot_data["borough"], rate, color="#7c9885")
ax.errorbar(rate, plot_data["borough"], xerr=1.96 * se, fmt="none", color="#1f2933")
ax.set(xlabel="Proportion closed in 2+ hours", ylabel="")
ax.set_xlim(left=0)
plt.show()
```

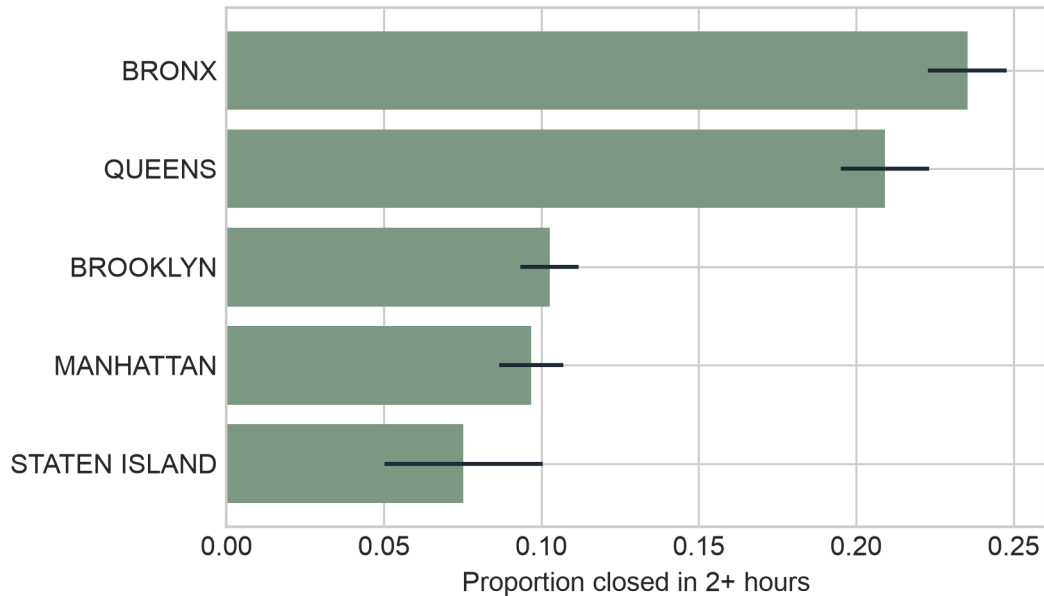


Figure 2: Share of complaints closed in two hours or more, by borough (95% normal-approximation intervals).

The Bronx has the largest observed two-hour rate (23.5%), while Staten Island has the smallest (7.5%). These differences are descriptive and are not adjusted estimates.

4 Group comparisons

Because closure times are strongly skewed, the analysis uses Kruskal-Wallis tests instead of one-way ANOVA. The chi-square tests assess whether the binary two-hour outcome is independent of each grouping variable. Cramer's V reports effect size, which is more informative than a p-value alone in a large sample.

```
tests = test_group_differences(complaints)
test_display = tests.copy()
test_display["statistic"] = test_display["statistic"].map(lambda value: f"{value:.2f}")
test_display["p_value"] = test_display["p_value"].map(
    lambda value: "<0.0001" if value < 0.0001 else f"{value:.4f}")
)
test_display["effect_size"] = test_display["effect_size"].map(
    lambda value: "-" if pd.isna(value) else f"{value:.3f}")
)
test_display
```

	variable	test	statistic	p_value	effect_size
0	borough	Kruskal-Wallis	693.61	<0.0001	-
1	borough	Chi-square	460.49	<0.0001	0.173
2	complaint_type	Kruskal-Wallis	119.40	<0.0001	-
3	complaint_type	Chi-square	102.72	<0.0001	0.082
4	day_of_week	Kruskal-Wallis	619.50	<0.0001	-
5	day_of_week	Chi-square	471.62	<0.0001	0.175

These are associational tests, not causal estimates. A small p-value indicates that the observed distributions differ, but it does not establish that borough or complaint type causes a slower closure.

5 Predictive model

A logistic regression predicts whether closure takes at least two hours. The features are available when a complaint is created: borough, complaint type, submission channel, hour, weekday/weekend status, and coordinates. The split is stratified and reproducible. Numeric features are median-imputed and scaled; categorical features are most-frequent-imputed and one-hot encoded. Class weights reduce the tendency to predict only the majority class.

```
model_result = fit_response_model(complaints, random_state=42)
pd.DataFrame(
    {
        "Metric": ["Accuracy", "Balanced accuracy", "Precision", "Recall", "F1", "ROC AUC"],
        "Value": [
            model_result.metrics["accuracy"],
            model_result.metrics["balanced_accuracy"],
            model_result.metrics["precision"],
            model_result.metrics["recall"],
            model_result.metrics["f1"],
            model_result.metrics["roc_auc"],
        ],
    },
).round(3)
```

	Metric	Value
0	Accuracy	0.684
1	Balanced accuracy	0.681

	Metric	Value
2	Precision	0.292
3	Recall	0.675
4	F1	0.408
5	ROC AUC	0.728

On the test set, the model's ROC AUC is 0.728 and its balanced accuracy is 0.681. Weighting the classes raises recall for two-hour cases to 67.5%, with lower precision as the tradeoff.

```
pd.DataFrame(
    model_result.confusion_matrix,
    index=["Actual <2 hours", "Actual 2+ hours"],
    columns=["Predicted <2 hours", "Predicted 2+ hours"],
)
```

	Predicted <2 hours	Predicted 2+ hours
Actual <2 hours	1773	811
Actual 2+ hours	161	335

```
fig, ax = plt.subplots()
ax.plot(model_result.fpr, model_result.tpr, color="#28666e", linewidth=2,
        label=f"Logistic regression (AUC = {model_result.metrics['roc_auc']:.2f})")
ax.plot([0, 1], [0, 1], color="#6b7280", linestyle="--", label="Random classifier")
ax.set(xlabel="False-positive rate", ylabel="True-positive rate", xlim=(0, 1), ylim=(0, 1))
ax.legend(frameon=False, loc="lower right")
plt.show()
```

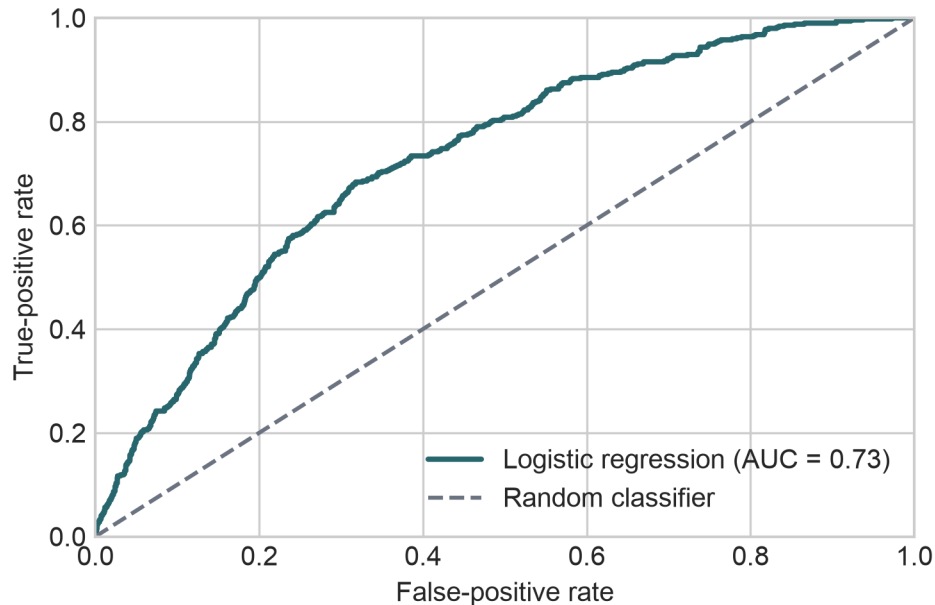


Figure 3: Receiver operating characteristic curve on the held-out test set.

```
pd.concat(
    [
        model_result.coefficients.nsmallest(6, "coefficient"),
```

```
model_result.coefficients.nlargest(6, "coefficient"),
]
).drop_duplicates().sort_values("coefficient").round(3)
```

	feature	coefficient
18	time_of_day_Evening	-0.771
4	complaint_type_Noise - House of Worship	-0.645
19	time_of_day_Overnight	-0.609
13	day_of_week_Saturday	-0.539
7	complaint_type_Noise - Street/Sidewalk	-0.454
1	borough_MANHATTAN	-0.433
14	day_of_week_Sunday	0.613
23	longitude	0.655
12	day_of_week_Monday	0.975
17	day_of_week_Wednesday	1.191
16	day_of_week_Tuesday	1.571
3	borough_STATEN ISLAND	1.912

Coefficients describe conditional associations on the log-odds scale. They should not be interpreted as causal effects, and the model should not be used to rank neighborhoods or allocate enforcement resources without a fairness and drift review.

6 Conclusions and limitations

The typical complaint closes in well under two hours, but the distribution has a long right tail. Two-hour closure rates vary substantially by borough, with the Bronx and Queens above the citywide sample rate and Staten Island, Manhattan, and Brooklyn below it. The group tests reject equal distributions, although their effect sizes are modest. The logistic model has useful but limited discrimination and should be treated as a descriptive baseline rather than a production system.

Important limitations are:

1. Closure time is an administrative measure; it is not the same as time to resolve the underlying noise problem.
2. The seven-day sample may reflect a holiday-week pattern and cannot establish long-run seasonality.
3. Complaint records measure reports to 311, not the true prevalence of noise.
4. NYC Open Data can be revised after publication; the committed snapshot is therefore the reproducibility reference for this report.
5. Unmeasured operational factors may explain observed geographic differences.

7 Reproducibility

The repository contains the fixed raw snapshot, data-download script, reusable analysis functions, dependency specification, tests, and rendering commands. Run `make test` to verify the transformations and `make render` to recreate this PDF. See `README.md` for complete setup instructions.